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DEPARTMENT OF ELECTRONICS



BIGPROJECT REPORT

**Thiết kế mạch đo nhiệt độ sử dụng
Thermistor**

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SUBJECT: Applied Electronics (EE3129)

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Thiết kế mạch đo nhiệt độ sử dụng Thermistor. Mạch có tầm đo $20 - 80^{\circ}\text{C}$, độ phân giải 0.1°C , ngõ ra là điện áp tương ứng từ $0 - 5\text{V}$:

1. Lựa chọn và mô phỏng Thermistor 2 dây (có điện trở dây dẫn là 10Ω).
2. Lựa chọn điện trở song song để tuyến tính hóa.
3. Lựa chọn OPAMP sử dụng, mô phỏng OPAMP theo datasheet.
4. Đảm bảo sai số (giữa giá trị lý thuyết và thực tế đo qua mạch) $\pm 5\%$.
5. Thiết kế mạch đọc điện áp cho ngõ ra ADC. Tiến hành lựa chọn ADC (số bit, nguồn).
Mô phỏng mạch đọc và kiểm chứng.

Chapter 1 Thực hiện mạch đọc điện áp cho ngõ ra ADC

1.1 Tổng quan về ADC

Theo chuẩn ngõ ra điện áp đọc từ Sensor là từ $0 \rightarrow 5\text{ V}$, với ngõ vào là từ $20^\circ\text{C} \rightarrow 80^\circ\text{C}$ với bước nhảy là 0.1°C . Thực hiện tính giá trị của một bước nhảy nhỏ nhất là:

$$\frac{5 - 0}{\frac{80 - 20}{0.1} + 1} \approx 8.3195\text{ mV}$$

Vậy có nghĩa là $\Delta_{min} \approx 8.3195\text{ mV}$, từ đó ta có:

$$2^Q = \frac{V_{ref+} - V_{ref-}}{\Delta_{min}}$$

Giả sử, chọn $V_{ref+} = 5\text{ V}$ và $V_{ref-} = 0\text{ V}$. Ta có $Q \approx 9.23\text{ bit}$

Tuy nhiên, trong thị trường không có ADC 9.23 bit cho nên nhóm em sẽ chọn con có trên thị trường là **ADS1115IDGSR**.

1.2 Electrical Characteristics của ADS1115

- Ultra-small packages:
 - + X2QFN: $2\text{ mm} \times 1.5\text{ mm} \times 0.4\text{ mm}$
 - + SOT: $2.9\text{ mm} \times 2.8\text{ mm} \times 0.6\text{ mm}$
- Wide supply range: 2.0 V to 5.5 V
- Low current consumption: $150\text{ }\mu\text{A}$ (continuous-conversion mode)
- Programmable data rate: 8SPS to 860SPS
- Single-cycle settling
- Internal low-drift voltage reference
- Internal oscillator
- I2C interface: four pin-selectable addresses

- Operating temperature range: -40°C to $+125^{\circ}\text{C}$
- Family of devices:
 - + ADS1115: four single-ended or two differential inputs with comparator and PGA

Từ những thông tin, ADC ADS1115 có tầm hoạt động ở dải nhiệt độ phù hợp trong khoảng $20^{\circ}\text{C} \rightarrow 80^{\circ}\text{C}$, nhóm em sử dụng nguồn cấp là $V_{DD} = 5.5\text{V}$ và cho phép nguồn sai số trong khoảng từ $[5.3, 5.5]\text{V}$ phù hợp với thông số dưới đây:

DIGITAL INPUT/OUTPUT					
V_{IH}	High-level input voltage		0.7 VDD	5.5	V
V_{IL}	Low-level input voltage		GND	0.3 VDD	V
V_{OL}	Low-level output voltage	$I_{OL} = 3\text{mA}$	GND	0.15	0.4 V
	Input leakage current	$GND < V_{DIG} < VDD$	-10	10	μA

Hình 1.1: Digital Input/Output của ADS1115.

Như vậy, với $GND \approx 0\text{V} < V_{IN} < V_{DD} = 5.5\text{V}$ thì có thể phù hợp cho tầm đo từ $0 \rightarrow 5\text{V}$ từ mạch đọc giá trị của sensor.

Tiếp theo, ta cần quan tâm đến giá trị như sau:

SYSTEM PERFORMANCE				
	Resolution (no missing codes)		16	Bits
DR	Data rate		8, 16, 32, 64, 128, 250, 475, 860	SPS
	Data rate variation	All data rates	-10% 10%	
	Output noise		See Noise Performance section	
INL	Integral nonlinearity	DR = 8SPS, FSR = $\pm 2.048V^{(2)}$	1	LSB
	Offset error	FSR = $\pm 2.048V$, differential inputs	3	LSB
		FSR = $\pm 2.048V$, single-ended inputs	± 3	LSB
	Offset drift over temperature	FSR = $\pm 2.048V$	0.005	LSB/ $^{\circ}C$
	Long-term Offset drift	FSR = $\pm 2.048V$, $T_A = 125^{\circ}C$, 1000 hours	± 1	LSB
	Offset power-supply rejection	FSR = $\pm 2.048V$, DC supply variation	1	LSB/V
	Offset channel match	Match between any two inputs	3	LSB
	Gain error ⁽³⁾	FSR = $\pm 2.048V$, $T_A = 25^{\circ}C$	0.01% 0.15%	
	Gain drift over temperature ⁽³⁾	FSR = $\pm 0.256V$	7	ppm/ $^{\circ}C$
		FSR = $\pm 2.048V$	5 40	
		FSR = $\pm 6.144V^{(1)}$	5	
	Long-term gain drift ⁽³⁾	FSR = $\pm 2.048V$, $T_A = 125^{\circ}C$, 1000 hours	$\pm 0.05\%$	
	Gain power-supply rejection		80	ppm/V
	Gain match ⁽³⁾	Match between any two gains	0.02% 0.1%	
	Gain channel match	Match between any two inputs	0.05% 0.1%	
CMRR	Common-mode rejection ratio	At DC, FSR = $\pm 0.256V$	105	dB
		At DC, FSR = $\pm 2.048V$	100	
		At DC, FSR = $\pm 6.144V^{(1)}$	90	
		$f_{CM} = 60Hz$, DR = 8SPS	105	
		$f_{CM} = 50Hz$, DR = 8SPS	105	

Hình 1.2: Electrical Characteristics của ADS1115.

- Data rate (DR) là giá trị dữ liệu Digital được đổi theo đơn vị thời gian. Với DR = 128 Sample per second là giá trị default của ADS1115, với thời gian cách nhau đọc dữ liệu khoảng tầm 8 ms.
- Integral Nonlinearity (INL) là thước đo độ lệch của đặc tuyến chuyển đổi thực tế so với đặc tuyến lý tưởng trong các bộ chuyển đổi ADC/DAC, sau khi đã loại bỏ sai số offset và gain. Với INL = 1 LSB.
- Offset error là sai số khi sử dụng chế độ Single-Ended với giá trị là ± 3 LSB.
- Gain drift over temperature là sai số của phép đo khi có ảnh hưởng của nhiệt độ với tại FSR = $\pm 6.144V$ thì Gain drift = $5ppm/^{\circ}C$ tức là mỗi $^{\circ}C$ sẽ gây sai số là 5 phần ngàn.
- Common-mode Rejection Ratio (CMRR) là thước đo khả năng của một bộ khuếch đại vi sai trong việc loại bỏ tín hiệu chung xuất hiện đồng thời và cùng pha trên cả hai đầu vào. Với CMRR = 90 dB.

1.3 Typical Characteristics của ADS1115

- Single-Ended Offset Error vs Temperature:

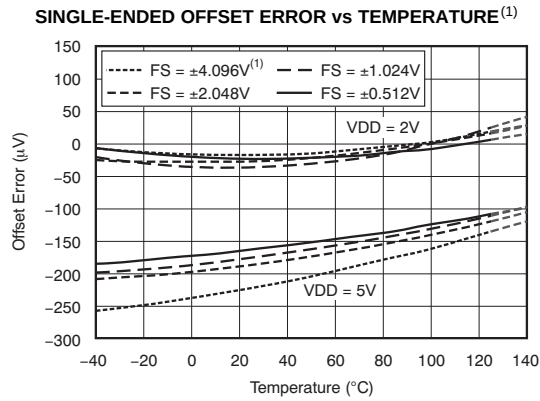


Figure 4.

+ Tại $T = 20^{\circ}\text{C}$: Offset Error = $-225\mu\text{V}$

+ Tại $T = 80^{\circ}\text{C}$: Offset Error = $-120\mu\text{V}$

- Gain Error vs Temperature:

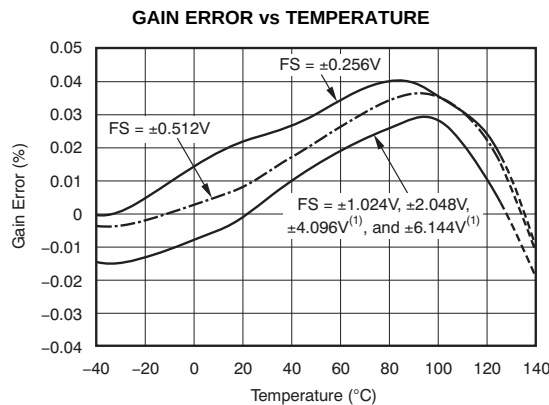


Figure 6.

+ Tại $T = 20^{\circ}\text{C}$: Gain Error = 0.00 %

+ Tại $T = 80^{\circ}\text{C}$: Gain Error = 0.03 %

- Gain Error vs Supply:

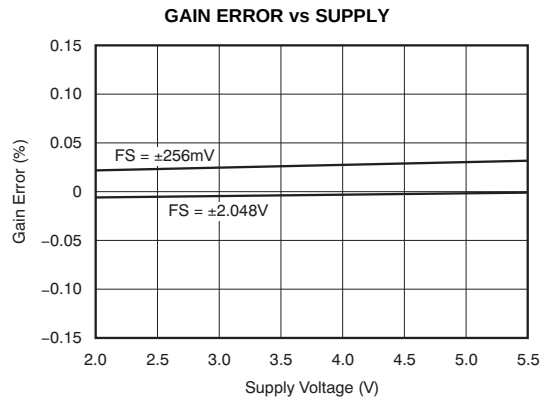


Figure 7.

+ Tại $V_{DD} = 5.5\text{V}$: Gain Error = 0.00 %

- INL vs Supply Voltage:

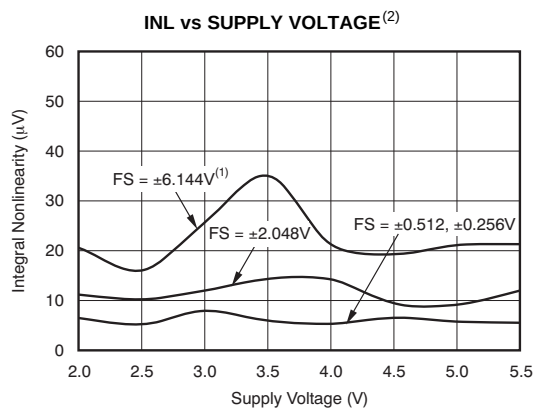


Figure 8.

+ Tại $V_{DD} = 5.5V$: INL vs Voltage = $20\mu V$

- INL vs Temperature:

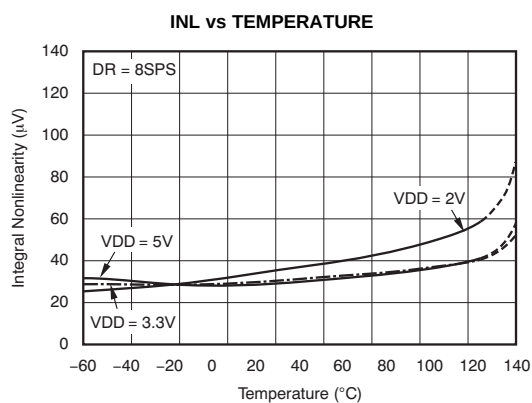


Figure 13.

+ Tại $T = 20^{\circ}C$: INL vs Temperature = $22\mu V$

+ Tại $T = 80^{\circ}C$: INL vs Temperature = $38\mu V$

- Data Rate vs Temperature:

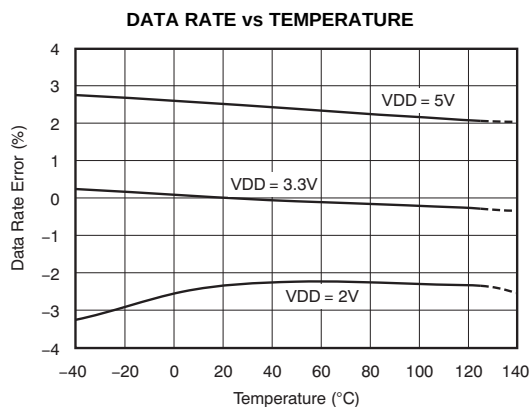


Figure 20.

+ Tại $T = 20^{\circ}C$: Data Rate Error = 2.8%

+ Tại $T = 80^{\circ}C$: Data Rate Error = 1.2%

1.4 Data Format

Với ADS1115 là một ADC 16-bit với ngõ ra là một chuỗi bit có dấu theo giá trị như sau:

Table 4. Input Signal versus Ideal Output Code

INPUT SIGNAL, V_{IN} ($A_{INP} - A_{INN}$)	IDEAL OUTPUT CODE ⁽¹⁾
$\geq FS (2^{15} - 1)/2^{15}$	7FFFh
$+FS/2^{15}$	0001h
0	0
$-FS/2^{15}$	FFFFh
$\leq -FS$	8000h

1. Excludes the effects of noise, INL, offset, and gain errors.

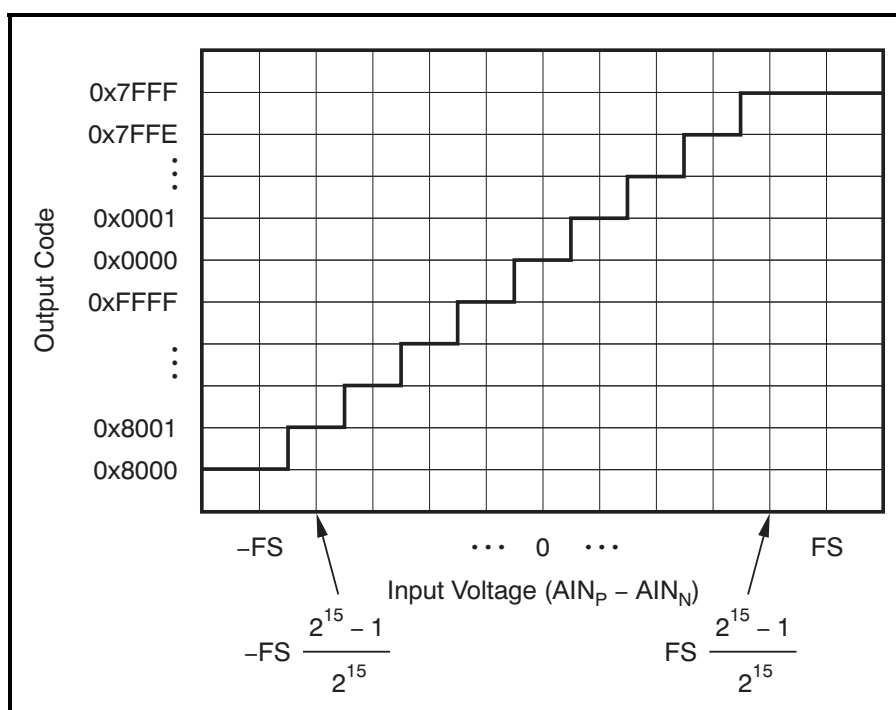


Figure 27. ADS1113/4/5 Code Transition Diagram

Hình 1.3: ADS1115 Code Transition Diagram.

Theo hình 1.3, với $FS = 6.144$ như ở trên thì giá trị lớn nhất mà ADC có thể đọc được

là $V_{ref} \times \frac{2^{15} - 1}{2^{15}} \approx 6.1438 \text{ V}$ vẫn phù hợp với tầm đo là từ $0 \rightarrow 5 \text{ V}$. Từ đó ta có công thức suy ngược từ D_{out} nhận được từ ADC là:

$$V_{out} = \frac{V_{ref}}{2^{15}} \times D_{out} = \frac{6.1438}{32768} \times D_{out}$$

1.5 Kết quả của việc tính toán ADC

Ta có bảng tính toán giá bên dưới, trong bảng giá trị có các giá trị như sau:

- **T_s** : là giá trị nhiệt độ chuẩn cần đo.
- **V_i** : là giá trị điện áp đo được từ sensor trong khoảng từ $0 \rightarrow 5 \text{ V}$ với sai số là 5%.
- **Offset_Error** : là giá trị sai số của ADS1115 do nhiệt độ tạo nên.
- **Gain_Error** : là giá trị của độ lợi của ADS1115 do nhiệt độ tạo nên.
- **V_i_ADC** : là giá trị mà ADC thực sự nhận tại chân IN0.
- **D_out** : là giá trị tính toán từ giá trị điện đọc từ chân IN0 với công thức là $\frac{V_{i_ADC} \times 2^{15}}{FS}$.
- **D_cal_Error** : là giá trị mà thực sự ADS1115 cho ra ở ngõ ra với các giá trị sai số cộng thêm do chính ADC và nhiệt độ.
- **Cal_V** : là giá trị được tính trong MCU chuyển đổi từ giá trị Digital của ADS1115 gửi về cho MCU.
- **Cal_Temp** : là giá trị chuyển đổi từ điện áp tính toán được Cal_V về giá trị nhiệt độ theo công thức chuẩn là $T = \frac{V_{in} + \frac{5}{3}}{\frac{1}{2}}$.

Requirement	
ERROR	5%

ADC					
	Min	Typ	Max		Unit
VDD	5.2		5.5	5.5	V
GND	-0.3		0	0.3	V
Resolution			16		Bit
Analog_In_V_to_GND					V
Temperature	-60		150		°C
PGA (program gain amplifier)	0.66666667		0.66666667		
FS (Full-Scale)	-6.142157353		6.1438125	6.142157353	V
Common-Mode Input Impedance			6000000		Ohm
DR (Data Rate)			128		SPS
Range_Temperature		20		80	°C
Offset Voltage	-0.000225			-0.00012	V
Gain Error	0%			0.03%	%
Gain Drift			0.000500%		%/°C
	Tai FS = 6.144		0.030000%		
Common Mode Rejection	Tai FS = 6.144		90		dB
INL vs. VoltageSupply		0.000020			V
INL vs. Temperature	0.000022		0.000038		V
Digital Input/Output					
V_IH	3.85		5.5		V
V_IL	-0.5		1.65		V

Calibration			
	Value_Standard	Value_Measurement	Error
raw_zero_25	0	0	0
raw_full_25	5	5	0
raw_zero_20	0	0	0
raw_full_20	5	5	0
raw_zero_50	0	0	0
raw_full_50	5	5	0
raw_zero_80	0	0	0
raw_full_80	5	5	0

Digital output			
Input Signal (V_I)	>= FS*(2 ¹⁵ -1)/(2 ¹⁵)		
Offset Error	-3	6.1419699091V	3 LSB
Offset Drift		0.005	LSB/°C
LSB	FS = 6.144V	0.3	LSB/V
Offset power supply rejection	VDD = 5.5V	0.3	LSB/V
INL		1	LSB

T_s	V_I	Offset_Vol	Gain_Error	S_E	V_I_ADC	Error_V_I_ADC	D_out	D_cal_error	IN	Cal_V	Cal_Temp	Error_re	Error_ab	
20	0.013320621	-0.000225	0.000000%	-0.001500	0.013583333	1.9722%	72.4684	74	0000000001001010	0.013874577	20.16649492	1%	3.6958%	0.166495
20.1	0.019954682	-0.000224	0.000100%	-0.001469	0.020207083	1.2649%	107.8067	106	000000000101010	0.019874393	20.23849272	1%	0.0086%	0.138493
20.2	0.025642402	-0.000224	0.000100%	-0.001439	0.025887233	0.9548%	138.1109	138	000000000100010	0.02587421	20.31049052	1%		0.110491
20.3	0.032330742	-0.000223	0.000200%	-0.001408	0.032565971	0.7276%	173.7426	177	000000000100001	0.033186487	20.39823785	0%		0.098238
20.4	0.038074328	-0.000223	0.000200%	-0.001378	0.038302627	0.5996%	204.3482	202	0000000001001010	0.037873844	20.45448613	0%		0.054486
20.5	0.044816929	-0.000222	0.000300%	-0.001347	0.045036343	0.4896%	240.2732	240	0000000001110000	0.044998627	20.53998352	0%		0.039984
20.6	0.05158625	-0.000222	0.000300%	-0.001317	0.051798108	0.4107%	276.3479	279	0000000001000111	0.052310904	20.62773084	0%		0.027331
20.7	0.058382283	-0.000221	0.000400%	-0.001286	0.058586016	0.3490%	312.562	316	0000000001011100	0.059248192	20.71097873	0%		0.010978
20.8	0.065205052	-0.000220	0.000400%	-0.001256	0.065400949	0.3005%	348.9204	347	0000000001010111	0.065060514	20.78072617	0%		0.019274
20.9	0.072054452	-0.000220	0.000500%	-0.001225	0.072244024	0.2631%	385.4288	387	0000000100000011	0.072560286	20.87072343	0%		0.029277
21	0.078930568	-0.000219	0.000500%	-0.001195	0.079113097	0.2313%	422.076	425	0000000101010011	0.079685068	20.95622082	0%		0.043779
21.1	0.084867512	-0.000219	0.000600%	-0.001164	0.085045587	0.2098%	453.7264	452	0000000110000100	0.084747414	21.01696896	0%		0.083031
21.2	0.091797781	-0.000218	0.000600%	-0.001134	0.091969553	0.1871%	490.6664	495	0000000111011111	0.092809668	21.11371601	0%		0.086284
21.3	0.098754705	-0.000218	0.000700%	-0.001103	0.098921658	0.1691%	527.7566	529	0000000100000001	0.099184473	21.19091368	1%		0.109786
21.4	0.105738272	-0.000217	0.000700%	-0.001073	0.105899704	0.1527%	564.9851	564	0000000100000100	0.105746773	21.26896127	1%		0.131039
21.5	0.112748471	-0.000216	0.000800%	-0.001042	0.112904885	0.1387%	602.3584	603	0000000100010101	0.11305905	21.3567086	1%		0.143291
21.6	0.118823527	-0.000216	0.000800%	-0.001012	0.118977181	0.1293%	634.7547	635	0000000100011011	0.119058867	21.4287064	1%		0.171294
21.7	0.125887774	-0.000215	0.000900%	-0.000981	0.126037188	0.1187%	672.4205	674	0000000101010010	0.126371143	21.51645372	1%		0.183546
21.8	0.132978615	-0.000215	0.000900%	-0.000951	0.133125068	0.1101%	710.235	714	000000010100010	0.133870914	21.60645097	1%		0.193549
21.9	0.140096038	-0.000214	0.001000%	-0.000920	0.140239071	0.1021%	748.1889	748	0000000110110100	0.14024572	21.68294864	1%		0.217051
22	0.146281565	-0.000214	0.001000%	-0.000890	0.1464233	0.0969%	781.1824	786	0000000100001010	0.147370502	21.76844603	1%		0.231554
22.1	0.153452935	-0.000213	0.001100%	-0.000859	0.153592049	0.0907%	819.4283	823	0000000110010111	0.154307791	21.85169349	1%		0.248307
22.2	0.160650843	-0.000212	0.001100%	-0.000829	0.160787599	0.0851%	857.8173	858	0000000101010110	0.160870079	21.93044109	1%		0.269559
22.3	0.166919306	-0.000212	0.001200%	-0.000798	0.167056042	0.0819%	891.26	892	0000000101011100	0.167244896	22.00693875	1%		0.293061
22.4	0.174171077	-0.000211	0.001200%	-0.000768	0.174306255	0.0776%	929.9406	934	0000000101000110	0.175119656	22.10143587	1%		0.298564
22.5	0.181449339	-0.000211	0.001300%	-0.000737	0.181584554	0.0745%	968.7711	969	0000000110000101	0.181681955	22.18018346	1%		0.319817
22.6	0.187800619	-0.000210	0.001300%	-0.000707	0.187935789	0.0720%	1002.6555	1006	0000000101010110	0.188619244	22.26343092	1%		0.336569
22.7	0.195132651	-0.000209	0.001400%	-0.000676	0.195267689	0.0692%	1041.7719	1041	0000010000000001	0.195181543	22.34217852	2%		0.357821
22.8	0.202491122	-0.000209	0.001400%	-0.000646	0.202627262	0.0672%	1081.0359	1080	0000010000010000	0.20249382	22.42992584	2%		0.370074
22.9	0.208925084	-0.000208	0.001500%	-0.000615	0.209062546	0.0658%	1115.3688	1116	0000000100000110	0.209243614	22.51092337	2%		0.389077
23	0.216337223	-0.000208	0.001500%	-0.000585	0.216476618	0.0644%	1154.9236	1157	0000000100000001	0.21693808	22.60317056	2%		0.396829
23.1	0.222826509	-0.000207	0.001600%	-0.000554	0.222998017	0.0635%	1189.5558	1189	0000000100010001	0.222930696	22.67516836	2%		0.424832
23.2	0.230292244	-0.000207	0.001600%	-0.000524	0.230436356	0.0627%	1229.401	1233	0000000100010100	0.231180445	22.77416534	2%		0.425685
23.3	0.238638788	-0.000206	0.001700%	-0.000493	0.238836393	0.0622%	1264.3323	1264	0000000100011000	0.238992767	22.84391321	2%		0.456087
23.4	0.244356043	-0.000205	0.001700%	-0.000463	0.244505864	0.0613%	1304.4623	1303	0000000100001001	0.244305044	22.93166053	2%		0.468339
23.5	0.251901602	-0.000205	0.001800%	-0.000432	0.252055739	0.0612%	1344.7416	1344	0000000101000000	0.25199231	23.02390771	2%		0.476092
23.6	0.258528469	-0.000204	0.001800%	-0.000402	0.25868965	0.0611%	1380.1173	1383	0000000100110011	0.259304586	23.1165504	2%		0.488345
23.7	0.266127425	-0.000204	0.001900%	-0.000371	0.266290652	0.0613%	1420.6862	1419	0000000100000101	0.26605438	23.19265257	2%		0.507347
23.8	0.272809362	-0.000203	0.001900%	-0.000341	0.272977297	0.0616%	1456.3601	1461	0000000101010101	0.27392914	23.28714968	2%		0.512850
23.9	0.28046163	-0.000203	0.002000%	-0.000310	0.280635651	0.0620%	1497.2182	1500	0000000101011010	0.281241417	23.374897	2%		0.525103
24	0.287198558	-0.000202	0.002000%	-0.000280	0.287378108	0.0625%	1533.1898	1534	0000000101111110	0.287616222	23.45139467	2%		0.548605
24.1	0.293963383	-0.000201	0.002100%	-0.000249	0.294149154	0.0632%	1569.314	1573	0000000100000100	0.29482989	23.53914199	2%		0.560858
24.2	0.300756084	-0.000201	0.002100%	-0.000219	0.300949189	0.0642%	1605.5929	1604	0000000100010000	0.300740822	23.60888986	2%		0.591110
24.3	0.307576642	-0.000200	0.002200%	-0.000188	0.307767971	0.0651%	1642.0188	1641	0000000100010001	0.30767811	23.69213732	3%		0.607863
24.4	0.315363091	-0.000200	0.002200%	-0.000158	0.31557124	0.0660%	1683.6029	1686	0000000100001010	0.316115353	23.79338423	2%		0.606616
24.5	0.322238425	-0.000199	0.002300%	-0.000127	0.32245448	0.0670%	1720.3257	1722	0000000101010110	0.322865147	23.87438176	3%		0.625618
24.6	0.329141548	-0.000199	0.002300%	-0.000097	0.329366005	0.0684%	1757.2025	1756	0000000101011011	0.329239952	23.95087943	3%		0.649121
24.7	0.337007873	-0.000198	0.002400%	-0.000066	0.337241619	0.0694%	1799.2165	1801	0000000100000100	0.337677195	24.05212634	3%		0.647874
24.8	0.34396563	-0.000197	0.002400%	-0.000036	0.344208241	0.0705%	1836.3841	1836	0000000100010100	0.344239494	24.13087393	3%		0.669126
24.9	0.350951106	-0.000197	0.002500%	-0.000005	0.351204351	0.0722%	1873.709	1878	0000000100010101	0.352114254	24.22537105	3%		0.674629
25	0.358897075	-0.000196	0.002500%	0.000025	0.359160052	0.0733%	1916.1534	1919	0000000101011111	0.359801519	24.31761823	3%		0.682382
25.1	0.365937033	-0.000196	0.002600%	0.000056	0.366211538	0.0750%	1953.7738	1954	0000000101000010	0.366363819	24.39636583	3%		0.703634
25.2	0.373004635	-0.000195	0.002600%	0.000086	0.373289733	0.0764%	1991.5366	1995	0000000101000011	0.374051085	24.48661301	3%		0.711387
25.3	0.381030001	-0.000195	0.002700%	0.000117	0.381327609	0.0781%	2034.4195	2034	0000000101010001	0.381363361	24.57636034	3%		0.723640
25.4	0.38815192	-0.000194	0.002700%	0.000147	0.388461019	0.0796%	2072.4769	2075	0000000100000001	0.389050627	24.66806752	3%		0.731392
25.5	0.395301412	-0.000193	0.002800%	0.000178	0.395622822	0.0813%	2110.6858	2114	0000000100000000	0.396362904	24.76535484	3%		0.743645
25.6	0.402478435	-0.000193	0.002800%	0.000208	0.402813208	0.0832%	2149.0472	2149	0000000100010001	0.402925203	24.83510244	3%		0.764698
25.7	0.410696553	-0.000192	0.002900%	0.000239	0.410957758	0.0848%	2192.4992	2193	0000000100000001	0.41079963	24.92599566	3%		0.770400
25.8	0.417840671	-0.000192	0.002900%	0.000269	0.418203152	0.0868%	2231.154	2236	0000000100010110	0.419237206	25.03084647	3%		0.791544
25.9	0.425092924	-0.000191	0.003000%	0.000300	0.425457865	0.0886%	2269.9546	2269	0000000100010101	0.425424517	25.1050942	3%		0.794096
26	0.432385229	-0.000191	0.003000%	0.000330	0.432777026	0.0906%	2308.507	2314	0000000100010000	0.433861759	25.2063411	3%		0.793659
26.1	0.43969861	-0.000190	0.003100%	0.000361	0.440105467	0.0925%	2348.005	2347	0000000100010101	0.44004907	25.28058884	3%		0.819411
26.2	0.447961445	-0.000189	0.003100%	0.000391	0.448383744	0.0943%	2392.1704	2393	0000000100010101	0.448673807	25.38408569	3%		0.815014
26.3	0.455328618	-0.000189	0.003200%	0.000422	0.455767927	0.0965%	2431.5657	2433	0000000100000001	0.456137578	2			

	28.4	0.614488409	-0.000177	0.004200%	0.001062	0.615376878	0.1446%	3283.0948	3286	0000100011010110	0.616106197	27.39327437	4%	1.006726
	28.5	0.622449859	-0.000176	0.004300%	0.001093	0.623365134	0.1470%	3325.7129	3330	0000110100000010	0.624355946	27.49227135	4%	1.007729
	28.6	0.630437792	-0.000176	0.004300%	0.001123	0.631380767	0.1496%	3368.4771	3372	0000110100010100	0.632230705	27.58676846	4%	1.013232
	28.7	0.637553119	-0.000175	0.004400%	0.001154	0.638522908	0.1521%	3406.5811	3405	0000110101000101	0.638418016	27.6610162	4%	1.038984
	28.8	0.645594828	-0.000174	0.004400%	0.001184	0.646592325	0.1545%	3449.6322	3453	0000110101111101	0.647417742	27.7690129	4%	1.030987
	28.9	0.653662689	-0.000174	0.004500%	0.001215	0.654690268	0.1572%	3492.8355	3492	0000110101000100	0.654730019	27.85676022	4%	1.043240
	29	0.661757259	-0.000173	0.004500%	0.001246	0.66281339	0.1598%	3536.1747	3541	0000110111010101	0.663817238	27.96700686	4%	1.032893
	29.1	0.669892628	-0.000173	0.004600%	0.001276	0.670968161	0.1623%	3574.8802	3577	0000110111110001	0.670867032	28.04800432	4%	1.051986
	29.2	0.677130426	-0.000172	0.004600%	0.001306	0.67824614	0.1648%	3618.5084	3622	0000110000100011	0.679104275	28.1492513	4%	1.050749
	29.3	0.685304307	-0.000172	0.004700%	0.001337	0.686452106	0.1675%	3662.298	3663	0000111001001111	0.68679154	28.24149848	4%	1.058502
	29.4	0.69261208	-0.000171	0.004700%	0.001367	0.693789411	0.1700%	3701.4332	3703	0000111001110111	0.694291311	28.33149574	4%	1.068504
	29.5	0.700839208	-0.000170	0.004800%	0.001398	0.702048991	0.1726%	3745.4967	3750	0000111010100110	0.703103542	28.43724251	4%	1.062757
	29.6	0.709092369	-0.000170	0.004800%	0.001428	0.710334644	0.1752%	3789.7036	3793	0000111011010001	0.711165796	28.53398956	4%	1.066010
	29.7	0.716481964	-0.000169	0.004900%	0.001459	0.717756086	0.1778%	3829.2977	3828	0000111011110100	0.717728096	28.61273715	4%	1.087263
	29.8	0.724788002	-0.000169	0.004900%	0.001489	0.726096073	0.1805%	3873.7924	3874	0000111100100010	0.726352833	28.71623399	4%	1.083766
	29.9	0.732232296	-0.000168	0.005000%	0.001520	0.733573239	0.1831%	3913.6838	3914	0000111101000101	0.733852604	28.80623125	4%	1.093769
	30	0.740591019	-0.000168	0.005000%	0.001550	0.741966978	0.1858%	3958.4652	3962	0000111101110110	0.742852329	28.91422795	4%	1.085772
	30.1	0.748089823	-0.000167	0.005100%	0.001581	0.74949969	0.1885%	3998.6529	4003	0000111101000011	0.750539595	29.00647514	4%	1.093525
	30.2	0.755616293	-0.000166	0.005100%	0.001611	0.757059823	0.1910%	4038.9869	4038	0000111111000010	0.757101894	29.08522273	4%	1.114777
	30.3	0.76405416	-0.000166	0.005200%	0.001642	0.765535073	0.1938%	4084.2032	4086	0000111111110110	0.76610162	29.19321944	4%	1.106781
	30.4	0.771634854	-0.000165	0.005200%	0.001672	0.773150464	0.1964%	4124.8321	4123	0001000000001011	0.773038908	29.2764669	4%	1.123533
	30.5	0.779243069	-0.000165	0.005300%	0.001703	0.780795666	0.1992%	4165.6199	4166	0001000000100010	0.781101162	29.37321394	4%	1.126786
	30.6	0.787759631	-0.000164	0.005300%	0.001733	0.789349473	0.2018%	4211.2553	4212	0001000000110100	0.789725899	29.47671078	4%	1.132389
	30.7	0.795421774	-0.000163	0.005400%	0.001764	0.797048665	0.2045%	4252.3313	4252	0001000000100110	0.79722567	29.56670804	4%	1.133292
	30.8	0.803900219	-0.000163	0.005400%	0.001794	0.805056463	0.2072%	4298.2547	4301	0001000000100101	0.806412889	29.67695467	4%	1.123045
	30.9	0.811706085	-0.000162	0.005500%	0.001825	0.813410453	0.2100%	4339.623	4338	0001000000110010	0.813350178	29.76020213	4%	1.139798
	31	0.820326202	-0.000162	0.005500%	0.001855	0.822071036	0.2127%	4385.828	4385	0001000010010001	0.822162409	29.86594891	4%	1.134051
	31.1	0.828095583	-0.000161	0.005600%	0.001886	0.829879626	0.2154%	4427.4876	4428	0001000010010100	0.830224663	29.96269595	4%	1.137304
	31.2	0.835892131	-0.000161	0.005600%	0.001916	0.837716084	0.2182%	4469.2959	4470	0001000010110110	0.838099422	30.05719307	4%	1.142807
	31.3	0.84458984	-0.000160	0.005700%	0.001947	0.846455777	0.2209%	4515.923	4521	0001000010101001	0.847661631	30.17193957	4%	1.128060
	31.4	0.852439586	-0.000159	0.005700%	0.001977	0.854345512	0.2236%	4558.0154	4560	0001000011010000	0.854973907	30.25968689	4%	1.140313
	31.5	0.860316339	-0.000159	0.005800%	0.002008	0.86228465	0.2263%	4600.2648	4605	0001000011111010	0.863441115	30.36095338	4%	1.139066
	31.6	0.868982248	-0.000158	0.005800%	0.002038	0.870355176	0.2291%	4651.9647	4651	0001001000101011	0.872036887	30.46443064	4%	1.135569
	31.7	0.877890886	-0.000157	0.005900%	0.002068	0.879927338	0.2320%	4694.4989	4699	0001001000101011	0.881035612	30.57242735	4%	1.127573
	31.8	0.885846367	-0.000157	0.005900%	0.002099	0.887925	0.2348%	4737.1852	4742	0001001010000110	0.889097866	30.66917439	4%	1.130826
	31.9	0.893828637	-0.000157	0.006000%	0.002130	0.895951891	0.2375%	4779.9895	4785	0001001010100001	0.89716012	30.76592144	4%	1.134079
	32	0.902704792	-0.000156	0.006000%	0.002160	0.904873193	0.2402%	4827.5855	4830	0001001011011110	0.905597363	30.86716835	4%	1.132832
	32.1	0.910739479	-0.000155	0.006100%	0.002191	0.912952631	0.2430%	4870.6901	4874	00010010100001010	0.913847111	30.96616533	4%	1.133835
	32.2	0.918800785	-0.000155	0.006100%	0.002221	0.921059329	0.2458%	4913.9401	4913	0001001010010001	0.921159388	31.05391265	4%	1.146087
	32.3	0.926888652	-0.000154	0.006200%	0.002252	0.929193064	0.2486%	4957.3343	4957	0001001010101101	0.929409136	31.15290963	4%	1.147090
	32.4	0.935003022	-0.000154	0.006200%	0.002282	0.937353931	0.2514%	5000.8733	5002	0001001100001010	0.937846378	31.25415654	4%	1.145843
	32.5	0.944006025	-0.000153	0.006300%	0.002313	0.946405927	0.2542%	5049.1666	5050	00010011011011010	0.946846104	31.36215324	4%	1.137847
	32.6	0.952172229	-0.000153	0.006300%	0.002343	0.954619768	0.2570%	5092.9882	5092	00010011110010010	0.954720863	31.45665036	4%	1.143350
	32.7	0.959504569	-0.000152	0.006400%	0.002374	0.961998215	0.2599%	5132.353	5137	00010100000010001	0.963158106	31.55789727	3%	1.142103
	32.8	0.967724366	-0.000151	0.006400%	0.002404	0.970265746	0.2626%	5176.461	5180	0001010000011100	0.97122036	31.65464432	3%	1.145356
	32.9	0.975970371	-0.000151	0.006500%	0.002435	0.978562028	0.2655%	5220.7225	5222	0001010000100110	0.979095119	31.74914143	3%	1.150859
	33	0.984242523	-0.000150	0.006500%	0.002465	0.986883047	0.2683%	5265.1159	5266	0001010010010010	0.987344868	31.84813841	3%	1.151862
	33.1	0.992540761	-0.000150	0.006600%	0.002496	0.995232717	0.2712%	5309.6622	5312	0001010010100000	0.995969604	31.95163525	3%	1.148365
	33.2	1.000865022	-0.000149	0.006600%	0.002526	1.003606887	0.2740%	5354.3398	5359	0001010011011011	1.004781836	32.05738203	3%	1.142618
	33.3	1.009215245	-0.000149	0.006700%	0.002557	1.012009086	0.2769%	5399.1696	5402	00010101000101010	1.01284409	32.15412907	3%	1.145871
	33.4	1.017591368	-0.000148	0.006700%	0.002587	1.020437089	0.2797%	5444.13	5446	0001010101000110	1.021093838	32.25312605	3%	1.146874
	33.5	1.025993327	-0.000147	0.006800%	0.002618	1.028691814	0.2825%	5489.2368	5491	0001010101100011	1.02953108	32.35437296	3%	1.146627
	33.6	1.034421058	-0.000147	0.006800%	0.002648	1.037372868	0.2854%	5534.484	5535	0001010100110111	1.037780828	32.45336994	3%	1.146630
	33.7	1.042874498	-0.000146	0.006900%	0.002679	1.045880256	0.2882%	5579.8717	5582	00010101100011010	1.04659306	32.55911671	3%	1.140883
	33.8	1.051353582	-0.000146	0.006900%	0.002709	1.054413832	0.2911%	5625.3992	5629	00010101111110101	1.055405291	32.66486349	3%	1.135137
	33.9	1.059858245	-0.000145	0.007000%	0.002740	1.062979334	0.2939%	5671.0665	5670	00010100001001010	1.063092556	32.75711067	3%	1.142889
	34	1.067541328	-0.000145	0.007000%	0.002770	1.070710035	0.2968%	5712.341	5711	00010101001001111	1.070779821	32.84935786	3%	1.150642
	34.1	1.076097968	-0.000144	0.007100%	0.002801	1.079322989	0.2997%	5758.2919	5757	00010101011101101	1.079404558	32.9528547	3%	1.147145
	34.2	1.084679988	-0.000143	0.007100%	0.002831	1.087960342	0.3025%	5804.3757	5808	0001010101010000	1.088966766	33.0676012	3%	1.132389
	34.3	1.093287325	-0.000143	0.007200%	0.002862	1.096626704	0.3054%	5850.6089	5845	00010101010101001	1.098654032	33.15984838	3%	1.140152
	34.4	1.101919908	-0.000142	0.007200%	0.002892	1.105316315	0.3082%	5896.9688	5901	000101100000101	1.106403734	33.27864481	3%	1.123155
	34.5	1.110577672	-0.000142	0.007300%	0.002923	1.114033823	0.3112%	5943.4776	5943	00010110010101011	1.114278494	33.37134193	3%	1.128658
	34.6	1.118419569	-0.000141	0.007300%	0.002953	1.121931462	0.3140%	5985.6122	5987	0001011010100011	1.122528242	33.47033891	3%	1.129661
	34.7	1.127128509	-0.000140	0.007400%	0.002984	1.130700346	0.3169%	6032.3951	6036	00010111001001010	1.131715462	33.5805855		

	36.9	1.315110094	-0.000128	0.008500%	0.003655	1.320120924	0.3810%	7042.9721	7042	0001101110000010	1.320334705	35.84401646	3%	1.055984	
	37	1.324387549	-0.000127	0.008500%	0.003685	1.329471437	0.3839%	7092.8579	7093	0001101110110101	1.329896914	35.95876296	3%	1.041237	
	37.1	1.332873218	-0.000127	0.008600%	0.003716	1.338030123	0.3869%	7138.5194	7139	0001101111100011	1.33852185	36.06225988	3%	1.037440	
	37.2	1.340570175	-0.000126	0.008600%	0.003746	1.345795307	0.3898%	7179.9474	7180	0001110000000110	1.346208916	36.15450699	3%	1.045493	
	37.3	1.349107395	-0.000126	0.008700%	0.003777	1.35440677	0.3928%	7225.8903	7228	0001110000111100	1.355208641	36.26250369	3%	1.037496	
	37.4	1.358481641	-0.000125	0.008700%	0.003807	1.3638959523	0.3957%	7276.3062	7275	0001110001101011	1.364020872	36.36825947	3%	1.031750	
	37.5	1.367067984	-0.000124	0.008800%	0.003838	1.3727171863	0.3986%	7322.5128	7327	0001100100110111	1.373776575	36.48524689	3%	1.014763	
	37.6	1.375679321	-0.000124	0.008800%	0.003868	1.381203592	0.4015%	7368.854	7373	0001100110011001	1.382395311	36.58874374	3%	1.011256	
	37.7	1.384315544	-0.000123	0.008900%	0.003899	1.389015599	0.4045%	7415.3334	7414	0001100111101110	1.390082577	36.68090992	3%	1.019009	
	37.8	1.393784257	-0.000123	0.008900%	0.003929	1.398463522	0.4075%	7466.2724	7466	00011010010101010	1.399832279	36.79798735	3%	1.020213	
	37.9	1.402468973	-0.000122	0.009000%	0.003960	1.406225316	0.4104%	7513.0175	7512	0001101010101000	1.408457016	36.90148419	3%	0.998516	
	38	1.411178344	-0.000122	0.009000%	0.003990	1.417011199	0.4134%	7559.8952	7563	0001101100001011	1.418019224	37.01623069	3%	0.983769	
	38.1	1.419912286	-0.000121	0.009100%	0.004021	1.425824293	0.4164%	7606.9097	7606	0001101101101110	1.426081478	37.11297774	3%	0.987022	
	38.2	1.429474163	-0.000120	0.009100%	0.004051	1.435467005	0.4192%	7658.3545	7663	0001101111101111	1.436768652	37.24122382	3%	0.958776	
	38.3	1.438255946	-0.000120	0.009200%	0.004082	1.444329462	0.4223%	7705.6365	7704	0001110000011000	1.444455917	37.33347101	3%	0.986529	
	38.4	1.447062055	-0.000119	0.009200%	0.004112	1.453214435	0.4252%	7753.0387	7758	00011110010001100	1.45420562	37.45046744	2%	0.949533	
	38.5	1.455892409	-0.000119	0.009300%	0.004143	1.46212676	0.4282%	7800.5868	7800	0001111001110000	1.462455368	37.54946442	2%	0.950536	
	38.6	1.464746923	-0.000118	0.009300%	0.004173	1.47106142	0.4311%	7848.2541	7848	0001111010101000	1.471455093	37.65746112	2%	0.942539	
	38.7	1.472827387	-0.000117	0.009400%	0.004204	1.479220788	0.4341%	7891.7851	7895	00011110110101011	1.480267324	37.76320789	2%	0.936792	
	38.8	1.481731037	-0.000117	0.009400%	0.004234	1.488206859	0.4370%	7939.7267	7943	0001111100000111	1.48926705	37.8712046	2%	0.928795	
	38.9	1.490658597	-0.000116	0.009500%	0.004265	1.497217984	0.4400%	7987.8019	7988	00011111001100100	1.497704292	37.97245151	2%	0.927548	
	39	1.499609982	-0.000116	0.009500%	0.004295	1.50625309	0.4430%	8036.0051	8036	0001111101100100	1.506704018	38.08044821	2%	0.919552	
	39.1	1.508585107	-0.000115	0.009600%	0.004326	1.515313101	0.4460%	8084.3411	8087	0001111110010111	1.516266226	38.19519471	2%	0.940805	
	39.2	1.516791096	-0.000115	0.009600%	0.004356	1.523600048	0.4489%	8128.5559	8130	0001111111000010	1.52432848	38.29194176	2%	0.908058	
	39.3	1.525814514	-0.000114	0.009700%	0.004387	1.532710259	0.4519%	8177.1566	8176	00011111111110000	1.532953217	38.3954386	2%	0.904561	
	39.4	1.534070765	-0.000113	0.009700%	0.004417	1.54104832	0.4548%	8221.641	8222	0010000000001110	1.541577953	38.49893544	2%	0.901065	
	39.5	1.543142133	-0.000113	0.009800%	0.004448	1.550208201	0.4579%	8270.5098	8271	0010000001001111	1.550765173	38.60918208	2%	0.890818	
	39.6	1.55223681	-0.000112	0.009800%	0.004478	1.559389496	0.4608%	8319.4929	8322	0010000010000010	1.560327381	38.72392857	2%	0.876071	
	39.7	1.56135471	-0.000112	0.009900%	0.004509	1.568597253	0.4639%	8368.6172	8371	0010000010110011	1.569514601	38.83417521	2%	0.865825	
	39.8	1.569709386	-0.000111	0.009900%	0.004539	1.577036291	0.4668%	8413.6402	8415	0010000011011111	1.577764349	38.93317219	2%	0.866828	
	39.9	1.578787454	-0.000111	0.010000%	0.004570	1.586296233	0.4698%	8463.0237	8467	0010000100010011	1.587514051	39.05016862	2%	0.849831	
	40	1.588062653	-0.000110	0.010000%	0.004600	1.595709009	0.4727%	8512.5194	8511	0010000100111111	1.59578738	39.14916556	2%	0.850584	
	40.1	1.598499504	-0.000109	0.010100%	0.004631	1.604806757	0.4757%	8567.9517	8568	0010000101101110	1.604527631	39.26491237	2%	0.845088	
	40.2	1.605725342	-0.000109	0.010100%	0.004661	1.613412188	0.4787%	8607.7991	8612	0010000110010010	1.614700722	39.37640866	2%	0.823591	
	40.3	1.614968287	-0.000108	0.010200%	0.004692	1.622763602	0.4817%	8657.5945	8660	0010000110101010	1.623700447	39.48440536	2%	0.815595	
	40.4	1.623483103	-0.000108	0.010200%	0.004722	1.631352073	0.4847%	8703.4202	8706	0010001000000010	1.632325184	39.58790221	2%	0.812098	
	40.5	1.632788822	-0.000107	0.010300%	0.004753	1.640750031	0.4877%	8753.5592	8758	0010001000110110	1.642074896	39.70489864	2%	0.795101	
	40.6	1.641335221	-0.000107	0.010300%	0.004783	1.649398971	0.4907%	8799.6487	8804	0010001001100100	1.650699623	39.80839548	2%	0.791605	
	40.7	1.650848768	-0.000106	0.010400%	0.004814	1.658834117	0.4937%	8850.0395	8854	0010001001001110	1.660074337	39.92089204	2%	0.779108	
	40.8	1.659280977	-0.000105	0.010400%	0.004844	1.667521168	0.4966%	8896.3858	8900	0010001010001000	1.668699074	40.02438889	2%	0.775611	
	40.9	1.668675989	-0.000105	0.010500%	0.004875	1.677014151	0.4997%	8947.0317	8946	00100010111001010	1.677323811	40.12788573	2%	0.772114	
	41	1.677319644	-0.000104	0.010500%	0.004905	1.685749963	0.5026%	8993.6381	8997	0010001100100101	1.686886019	40.24263223	2%	0.757398	
	41.1	1.686759755	-0.000104	0.010600%	0.004936	1.695289418	0.5057%	9044.532	9046	0010001101010110	1.696073238	40.35287888	2%	0.747121	
	41.2	1.694679247	-0.000103	0.010600%	0.004966	1.70329854	0.5086%	9087.2615	9087	0010001101111111	1.703760504	40.44512605	2%	0.754874	
	41.3	1.703395014	-0.000103	0.010700%	0.004997	1.712111143	0.5117%	9134.2776	9135	0010001101011111	1.712760229	40.55312275	2%	0.746877	
	41.4	1.712903706	-0.000102	0.010700%	0.005027	1.721718493	0.5146%	9185.5337	9190	0010001111100110	1.723072414	40.67686897	2%	0.723131	
	41.5	1.721666002	-0.000101	0.010800%	0.005058	1.730577972	0.5176%	9232.7999	9231	0010010000001111	1.73075968	40.76911616	2%	0.730884	
	41.6	1.730451967	-0.000101	0.010800%	0.005088	1.73946105	0.5206%	9280.1919	9281	0010010001000001	1.740134394	40.88161272	2%	0.718387	
	41.7	1.740027319	-0.000100	0.010900%	0.005119	1.749138864	0.5236%	9331.824	9334	0010010001101110	1.75007159	41.00085909	2%	0.699141	
	41.8	1.748859255	-0.000100	0.010900%	0.005149	1.758069255	0.5266%	9379.4685	9383	0010010010100111	1.75925881	41.11110572	2%	0.688894	
	41.9	1.758478215	-0.000099	0.011000%	0.005180	1.767792077	0.5297%	9431.3407	9436	00100100101101100	1.769196007	41.23035208	2%	0.669648	
	42	1.767355744	-0.000099	0.011000%	0.005210	1.77676941	0.5326%	9479.2356	9484	0010010100001100	1.778195732	41.33834879	2%	0.661651	
	42.1	1.776256472	-0.000098	0.011100%	0.005241	1.785771396	0.5357%	9527.2621	9527	0010010101010111	1.786257986	41.43509583	2%	0.664904	
	42.2	1.785940674	-0.000097	0.011100%	0.005271	1.795559764	0.5386%	9579.484	9583	0010010101101111	1.796757666	41.56109199	2%	0.638908	
	42.3	1.794886427	-0.000097	0.011200%	0.005302	1.804609157	0.5417%	9627.7634	9628	00100101001101010	1.80481992	41.65783904	2%	0.642161	
	42.4	1.803855094	-0.000096	0.011200%	0.005332	1.813679323	0.5446%	9676.1536	9678	0010010110010010	1.814194633	41.7703356	1%	0.629664	
	42.5	1.812846581	-0.000096	0.011300%	0.005363	1.822775817	0.5477%	9724.6842	9724	0010010111111100	1.823194359	41.87833231	1%	0.621668	
	42.6	1.821860791	-0.000095	0.011300%	0.005393	1.831892876	0.5507%	9773.3246	9773	0010010000101101	1.832361578	41.98857694	1%	0.611421	
	42.7	1.830897626	-0.000094	0.011400%	0.005424	1.841035096	0.5537%	9822.0992	9824	0010010010010000	1.841943787	42.10332544	1%	0.596675	
	42.8	1.839856993	-0.000094	0.011400%	0.005454	1.850199661	0.5567%	9870.9931	9874	00100101001001010	1.851318591	42.21582201	1%	0.584178	
	42.9	1.848038792	-0.000093	0.011500%	0.005485	1.859388218	0.5597%	9920.0149	9921	0010010110000001	1.860130732	42.32156878	1%	0.578431	
	43	1.857391304	-0.000093	0.011500%	0.005515	1.867843139	0.5627%	9965.1227	9968	0010010111100000	1.869842963	42.42731555	1%	0.572684	
	43.1	1.866518773	-0.000092	0.011600%	0.005546	1.87707873	0.5658%	10014.3955							

	45.4	2.076557867	-0.000079	0.012700%	0.006247	2.089749168	0.6352%	11149.0127	11152	0010101110010000	2.090936188	45.09123425	1%	0.308766
	45.5	2.085473762	-0.000078	0.012800%	0.006278	2.098785573	0.6383%	11197.2228	11196	0010101110111100	2.099185936	45.19023123	1%	0.309769
	45.6	2.095134571	-0.000078	0.012800%	0.006308	2.108571072	0.6413%	11249.4294	11251	0010101111110011	2.109498121	45.31397746	1%	0.286023
	45.7	2.104093093	-0.000077	0.012900%	0.006339	2.117651519	0.6444%	11297.8475	11297	0010110000100001	2.118122858	45.41174743	1%	0.282526
	45.8	2.113073329	-0.000077	0.012900%	0.006369	2.126753248	0.6474%	11346.4331	11348	0010110000101001	2.127685066	45.53222079	1%	0.286779
	45.9	2.122794843	-0.000076	0.013000%	0.006400	2.136905279	0.6505%	11396.9981	11400	0010110010010001	2.137434799	45.64821722	1%	0.250783
	46	2.131817076	-0.000076	0.013000%	0.006429	2.146747962	0.6535%	11447.7712	11446	0010110010111000	2.146434484	45.75721393	1%	0.242786
	46.1	2.140869705	-0.000075	0.013100%	0.006461	2.154916245	0.6565%	11496.6854	11498	0010110011010101	2.155899209	45.86971049	0%	0.230290
	46.2	2.149925626	-0.000074	0.013100%	0.006491	2.164104452	0.6595%	11545.7053	11545	0010110000100101	2.164621439	45.97545727	0%	0.225443
	46.3	2.159011732	-0.000074	0.013200%	0.006522	2.173317732	0.6626%	11594.8591	11598	0010110101001110	2.174558636	46.09470363	0%	0.205296
	46.4	2.167404779	-0.000073	0.013200%	0.006552	2.181830754	0.6656%	11640.2769	11643	0010110101110111	2.182995878	46.19595054	0%	0.204049
	46.5	2.177247078	-0.000073	0.013300%	0.006583	2.191906358	0.6687%	11693.4977	11692	0010110110101010	2.192183098	46.30619717	0%	0.193803
	46.6	2.186396106	-0.000072	0.013300%	0.006613	2.201081474	0.6717%	11742.9813	11743	0010110111011111	2.201745306	46.42094367	0%	0.179056
	46.7	2.195565895	-0.000071	0.013400%	0.006644	2.210380277	0.6747%	11792.5913	11794	0010111000010010	2.211307514	46.53569017	0%	0.164310
	46.8	2.20475634	-0.000071	0.013400%	0.006674	2.219699225	0.6778%	11842.3088	11846	0010111001000110	2.221057217	46.6526866	0%	0.147313
	46.9	2.213967332	-0.000070	0.013500%	0.006705	2.229040068	0.6808%	11892.1462	11897	0010111001111001	2.230619425	46.76743331	0%	0.132567
	47	2.223198765	-0.000070	0.013500%	0.006735	2.238402012	0.6838%	11942.0899	11943	0010111010100111	2.239244162	46.87092994	0%	0.129070
	47.1	2.232450531	-0.000069	0.013600%	0.006766	2.247785657	0.6869%	11992.1526	11992	0010111011011000	2.248431381	46.98117657	0%	0.118823
	47.2	2.241722524	-0.000069	0.013600%	0.006796	2.257188896	0.6899%	12042.3203	12046	0010111100001110	2.258556072	47.10267287	0%	0.097327
	47.3	2.250310476	-0.000068	0.013700%	0.006827	2.265905416	0.6930%	12088.8233	12091	0010111100111011	2.266983315	47.20391798	0%	0.096080
	47.4	2.259623707	-0.000067	0.013700%	0.006857	2.275305387	0.6960%	12139.2131	12139	0010111101101011	2.27599304	47.31191648	0%	0.088084
	47.5	2.268956881	-0.000067	0.013800%	0.006888	2.284819277	0.6991%	12189.7305	12193	0010111101000011	2.286117731	47.43341277	0%	0.066587
	47.6	2.278309769	-0.000066	0.013800%	0.006918	2.294305373	0.7021%	12240.3398	12243	0010111110100111	2.295492445	47.54590934	0%	0.054091
	47.7	2.288382104	-0.000066	0.013900%	0.006949	2.304510981	0.7052%	12294.8348	12295	0011000000000011	2.305242147	47.66290577	0%	0.037094
	47.8	2.297074574	-0.000065	0.013900%	0.006979	2.313341983	0.7082%	12341.9019	12343	0011000000110111	2.314241873	47.77909247	0%	0.029098
	47.9	2.306486233	-0.000065	0.014000%	0.007010	2.322892301	0.7113%	12392.8538	12397	0011000001101011	2.324366564	47.89239877	0%	0.007601
	48	2.315917251	-0.000064	0.014000%	0.007040	2.332459357	0.7143%	12443.8949	12443	0011000010010101	2.332991301	47.99589561	0%	0.004104
	48.1	2.32467224	-0.000063	0.014100%	0.007071	2.341348505	0.7174%	12491.3194	12493	0011000011001011	2.342366014	48.10839217	0%	0.000392
	48.2	2.33344859	-0.000063	0.014100%	0.007101	2.350258363	0.7204%	12538.8543	12538	0011000011110101	2.350803257	48.20963908	0%	0.009639
	48.3	2.34293925	-0.000062	0.014200%	0.007132	2.359889568	0.7235%	12590.2377	12591	0011000100101111	2.360740454	48.32888544	0%	0.028885
	48.4	2.351756885	-0.000062	0.014200%	0.007162	2.368842116	0.7265%	12638.0004	12639	0011000101011111	2.369740179	48.43688215	0%	0.036882
	48.5	2.361286386	-0.000061	0.014300%	0.007193	2.378513618	0.7296%	12689.5988	12693	0011000110010101	2.37966487	48.55837844	0%	0.058378
	48.6	2.370144465	-0.000061	0.014300%	0.007223	2.387976942	0.7326%	12737.5857	12740	0011000111000100	2.388612521	48.68412521	0%	0.064125
	48.7	2.379712769	-0.000060	0.014400%	0.007254	2.397219781	0.7357%	12789.398	12792	0011000111110000	2.398426804	48.78112164	0%	0.061122
	48.8	2.388611651	-0.000059	0.014400%	0.007284	2.406258393	0.7387%	12837.6039	12837	0011001000100011	2.406864046	48.88236855	0%	0.082369
	48.9	2.398217515	-0.000059	0.014500%	0.007315	2.416007184	0.7418%	12889.6306	12890	0011001001010101	2.416801243	49.00161491	0%	0.101615
	49	2.407156358	-0.000058	0.014500%	0.007345	2.425084176	0.7448%	12938.0572	12941	0011001100001011	2.426363451	49.11636141	0%	0.116361
	49.1	2.416799739	-0.000058	0.014600%	0.007376	2.434874949	0.7479%	12990.2392	12992	0011001101000000	2.435925659	49.23110791	0%	0.131108
	49.2	2.425778099	-0.000057	0.014600%	0.007406	2.443992881	0.7509%	13038.937	13043	0011001101110011	2.445487967	49.34585441	0%	0.145854
	49.3	2.434776606	-0.000057	0.014700%	0.007437	2.453135141	0.7540%	13087.7118	13090	0011001100100010	2.454300098	49.45160118	0%	0.151601
	49.4	2.444475984	-0.000056	0.014700%	0.007467	2.462980628	0.7570%	13140.2385	13145	0011001101010101	2.464612284	49.5753474	0%	0.175347
	49.5	2.45351334	-0.000055	0.014800%	0.007498	2.472162177	0.7601%	13189.2229	13192	0011001110001000	2.473424515	49.68109418	0%	0.181094
	49.6	2.462570509	-0.000055	0.014800%	0.007528	2.481362771	0.7631%	13238.309	13243	0011001101101101	2.482986723	49.79584068	0%	0.195841
	49.7	2.470969877	-0.000054	0.014900%	0.007559	2.489902613	0.7662%	13283.8698	13284	0011001111100100	2.490673988	49.88808768	0%	0.188088
	49.8	2.480743835	-0.000054	0.014900%	0.007589	2.499826578	0.7692%	13336.8151	13336	0011010000011000	2.500423691	50.00508429	0%	0.250084
	49.9	2.490535048	-0.000053	0.015000%	0.007620	2.509770054	0.7723%	13389.8645	13394	0011010001010010	2.511298359	50.13558031	0%	0.235580
	50	2.499669234	-0.000053	0.015000%	0.007650	2.519050526	0.7754%	13439.3768	13442	0011010010000010	2.520298084	50.24357701	0%	0.243577
	50.1	2.508822671	-0.000052	0.015100%	0.007681	2.528362501	0.7784%	13489.0037	13488	0011010010110000	2.528922821	50.34707385	0%	0.247074
	50.2	2.517995247	-0.000051	0.015100%	0.007711	2.537671786	0.7814%	13538.723	13541	0011010011010011	2.538860018	50.46632021	1%	0.266320
	50.3	2.527186848	-0.000051	0.015200%	0.007742	2.547014384	0.7846%	13588.5666	13593	0011010100011001	2.54860972	50.58331664	1%	0.283317
	50.4	2.536397362	-0.000050	0.015200%	0.007772	2.556373043	0.7876%	13638.496	13639	0011010101000111	2.557234457	50.68681348	1%	0.286813
	50.5	2.545626676	-0.000050	0.015300%	0.007803	2.565754808	0.7907%	13688.5488	13687	0011010101101111	2.566234182	50.79481019	1%	0.294810
	50.6	2.554874677	-0.000049	0.015300%	0.007833	2.575152391	0.7937%	13738.6856	13738	0011010101010101	2.575796391	50.90955669	1%	0.309557
	50.7	2.56414125	-0.000048	0.015400%	0.007864	2.584571867	0.7968%	13788.9394	13794	0011010111100010	2.58629607	51.03555284	1%	0.335553
	50.8	2.573426283	-0.000048	0.015400%	0.007894	2.594008932	0.7998%	13839.2871	13844	0011011000010100	2.595670784	51.14804941	1%	0.348049
	50.9	2.582729662	-0.000047	0.015500%	0.007925	2.603466676	0.8029%	13889.745	13889	0011011001000001	2.604108027	51.24929632	1%	0.349296
	51	2.592051274	-0.000047	0.015500%	0.007955	2.612941764	0.8059%	13940.2955	13939	0011011001110011	2.61348274	51.36179288	1%	0.361793
	51.1	2.601391004	-0.000046	0.015600%	0.007986	2.622437325	0.8090%	13990.9553	13995	0011011010101011	2.62396242	51.48778904	1%	0.387789
	51.2	2.610678951	-0.000046	0.015600%	0.008016	2.631983797	0.8121%	14038.152	14037	0011011011010101	2.633828516	51.58228516	1%	0.382286
	51.3	2.618804888	-0.000045	0.015700%	0.008047	2.640152731	0.8152%	14085.4697	14086	0011011100000110	2.641044399	51.69253279	1%	0.392533
	51.4	2.628209508	-0.000044	0.015700%	0.008077	2.649703996	0.8182%	14136.424	14139	0011011100110101	2.650961596	51.81177915	1%	0.411779
	51.5	2.636956271	-0.000044	0.015800%	0.008108	2.658613842	0.8213%	14183.9605	14184	0011011101010000	2.659418839	51.91302606	1%	0.413026
	51.6	2.646388209	-0.000043	0.015800%	0.008138	2.668202969	0.8243%	14235.1172	14239	0011011100111111	2.669731024	52.03677229	1%	0.436772
	51.7	2.655837579	-0.000043											

	53.9	2.858437073	-0.000030	0.017000%	0.008840	2.884019291	0.8950%	15386.5202	15389	0011110000011101	2.885349443	54.62419332	1%	0.724193
	54	2.867642315	-0.000030	0.017000%	0.008870	2.893394207	0.8980%	15436.5363	15440	0011110001010000	2.894911652	54.73893982	1%	0.738940
	54.1	2.876864423	-0.000029	0.017100%	0.008901	2.902788628	0.9011%	15486.6564	15487	0011110001111111	2.903723883	54.84468659	1%	0.744687
	54.2	2.88610328	-0.000028	0.017100%	0.008931	2.912197599	0.9041%	15536.8542	15541	0011110010110101	2.913848574	54.96618288	1%	0.766183
	54.3	2.894105618	-0.000028	0.017200%	0.008962	2.920363395	0.9073%	15580.4195	15579	0011110011011011	2.920973356	55.05168027	1%	0.751680
	54.4	2.903379823	-0.000027	0.017200%	0.008992	2.929809145	0.9103%	15630.8135	15631	0011110100001111	2.930723059	55.1686767	1%	0.768677
	54.5	2.9126730425	-0.000027	0.017300%	0.009023	2.939379589	0.9134%	15681.32	15685	0011110101000101	2.94064775	55.290173	1%	0.790173
	54.6	2.921977308	-0.000026	0.017300%	0.009053	2.948795987	0.9165%	15731.8861	15734	0011110101101110	2.950034669	55.40041963	1%	0.800430
	54.7	2.930678179	-0.000025	0.017400%	0.009084	2.957627742	0.9196%	15779.2284	15779	0011110110100011	2.958472212	55.50169654	1%	0.801967
	54.8	2.940018371	-0.000025	0.017400%	0.009114	2.967143387	0.9226%	15829.9853	15834	0011110110110110	2.968784397	55.62541277	2%	0.825413
	54.9	2.949374483	-0.000024	0.017500%	0.009145	2.97667768	0.9257%	15880.8616	15881	0011110000010001	2.977596628	55.73115954	2%	0.831160
	55	2.958746427	-0.000024	0.017500%	0.009175	2.986226523	0.9288%	15931.8056	15932	0011110001111000	2.987158836	55.84590604	2%	0.845906
	55.1	2.968134057	-0.000023	0.017600%	0.009206	2.995793801	0.9319%	15982.848	15984	0011110011100000	2.996908539	55.96290247	2%	0.862902
	55.2	2.976920574	-0.000023	0.017600%	0.009236	3.004752937	0.9349%	16030.6458	16034	0011110101000010	3.006283253	56.07539903	2%	0.875399
	55.3	2.98634034	-0.000022	0.017700%	0.009267	3.01435382	0.9381%	16081.8675	16085	0011110110101011	3.015845461	56.19014553	2%	0.890146
	55.4	2.995160951	-0.000021	0.017700%	0.009297	3.023347473	0.9411%	16129.8495	16130	0011111000000100	3.024282703	56.29139244	2%	0.891392
	55.5	3.004612383	-0.000021	0.017800%	0.009328	3.032982503	0.9442%	16181.2533	16180	0011111001101000	3.033657417	56.40388901	2%	0.903889
	55.6	3.013466616	-0.000020	0.017800%	0.009358	3.042011214	0.9472%	16229.4223	16232	0011111011010100	3.04340712	56.52088544	2%	0.920885
	55.7	3.022949244	-0.000020	0.017900%	0.009389	3.051678913	0.9504%	16281.0004	16285	0011111001101101	3.053344316	56.6401318	2%	0.940132
	55.8	3.031226505	-0.000019	0.017900%	0.009419	3.060126288	0.9534%	16326.068	16326	0011111100001100	3.061031582	56.73237898	2%	0.932379
	55.9	3.040131925	-0.000019	0.018000%	0.009450	3.069212435	0.9566%	16374.5434	16373	0011111111111011	3.069843813	56.83812576	2%	0.938126
	56	3.049054171	-0.000018	0.018000%	0.009480	3.078312019	0.9596%	16423.0906	16422	0100000000100110	3.079031033	56.94837239	2%	0.948372
	56.1	3.058599965	-0.000017	0.018100%	0.009511	3.088044808	0.9627%	16475.016	16479	0100000001011111	3.089718206	57.07661848	2%	0.976618
	56.2	3.067554415	-0.000017	0.018100%	0.009541	3.097179046	0.9657%	16523.748	16527	0100000010001111	3.098717932	57.18461518	2%	0.984615
	56.3	3.077129995	-0.000016	0.018200%	0.009572	3.106943099	0.9689%	16575.8402	16576	0100000010000000	3.107905151	57.29486182	2%	0.994862
	56.4	3.086116176	-0.000016	0.018200%	0.009602	3.116110512	0.9719%	16624.7482	16629	0100000011101011	3.117842348	57.41410818	2%	1.014108
	56.5	3.095118592	-0.000015	0.018300%	0.009633	3.125297019	0.9750%	16673.7601	16676	0100000100100100	3.126654579	57.51985495	2%	1.019855
	56.6	3.104137124	-0.000015	0.018300%	0.009663	3.134498219	0.9781%	16722.8494	16723	0100000101010011	3.13546681	57.62560172	2%	1.025602
	56.7	3.113171653	-0.000014	0.018400%	0.009694	3.143718296	0.9812%	16772.0394	16776	0100000110001000	3.145404007	57.74484808	2%	1.04848
	56.8	3.122222061	-0.000013	0.018400%	0.009724	3.1529518	0.9842%	16821.301	16824	0100000101110000	3.154403732	57.85284479	2%	1.052845
	56.9	3.13128823	-0.000013	0.018500%	0.009755	3.162205884	0.9874%	16870.673	16873	0100000111101001	3.163590952	57.96309142	2%	1.063091
	57	3.140379042	-0.000012	0.018500%	0.009785	3.171472731	0.9904%	16920.1098	16921	0100001000011101	3.172590677	58.07108813	2%	1.071088
	57.1	3.149467378	-0.000012	0.018600%	0.009816	3.18076914	0.9934%	16969.6558	16974	0100001001001110	3.182527874	58.18033448	2%	1.080334
	57.2	3.15858012	-0.000011	0.018600%	0.009846	3.190057898	0.9964%	17019.2654	17023	0100001001111111	3.191715094	58.30098112	2%	1.100981
	57.3	3.1677114368	-0.000011	0.018700%	0.009877	3.199777117	0.9997%	17065.7835	17067	0100001010010111	3.199964642	58.3995781	2%	1.099578
	57.4	3.176258652	-0.000010	0.018700%	0.009907	3.208108811	1.0028%	17115.569	17114	0100001010101010	3.208777073	58.50532487	2%	1.105325
	57.5	3.184826374	-0.000009	0.018800%	0.009938	3.216961972	1.0059%	17162.2679	17164	0100001100001000	3.218151787	58.61782144	2%	1.117821
	57.6	3.194592256	-0.000009	0.018800%	0.009968	3.226823656	1.0089%	17215.4145	17215	0100001100111111	3.227713995	58.73256794	2%	1.132568
	57.7	3.203781339	-0.000008	0.018900%	0.009999	3.236205579	1.0121%	17265.4679	17270	0100001101101100	3.238026818	58.85631416	2%	1.156314
	57.8	3.212396753	-0.000008	0.018900%	0.010029	3.24500634	1.0151%	17312.4208	17317	0100001101001011	3.246838411	58.96206094	2%	1.162061
	57.9	3.221616194	-0.000007	0.019000%	0.010060	3.254420077	1.0182%	17362.644	17361	0100001110100001	3.25508816	59.06105791	2%	1.161058
	58	3.230850096	-0.000007	0.019000%	0.010090	3.263846724	1.0213%	17412.9361	17416	0100010000001000	3.265400345	59.18480414	2%	1.184804
	58.1	3.239513216	-0.000006	0.019100%	0.010121	3.272699622	1.0244%	17460.1671	17462	0100010000110110	3.274025082	59.28830098	2%	1.188301
	58.2	3.248776766	-0.000005	0.019100%	0.010151	3.282156346	1.0275%	17510.6197	17514	0100010001101010	3.283747484	59.40529741	2%	1.205297
	58.3	3.257471456	-0.000005	0.019200%	0.010182	3.291043268	1.0306%	17558.0323	17560	0100010010011000	3.292399521	59.50879425	2%	1.206794
	58.4	3.266182291	-0.000004	0.019200%	0.010212	3.299942714	1.0336%	17605.5117	17608	0100010010100100	3.301399246	59.61679095	2%	1.216791
	58.5	3.275489964	-0.000004	0.019300%	0.010243	3.309450066	1.0368%	17656.2343	17656	0100010011111000	3.310398972	59.72478766	2%	1.224768
	58.6	3.284231654	-0.000003	0.019300%	0.010273	3.318381795	1.0398%	17703.8859	17706	0100010100101010	3.319773685	59.83728423	2%	1.237284
	58.7	3.294146453	-0.000002	0.019400%	0.010304	3.328502748	1.0429%	17757.8822	17761	0100010101000011	3.330085871	59.96103045	2%	1.261030
	58.8	3.302339864	-0.000002	0.019400%	0.010334	3.336882609	1.0460%	17802.5896	17803	0100010110001011	3.33796063	60.05552757	2%	1.265528
	58.9	3.311277485	-0.000001	0.019500%	0.010365	3.345865783	1.0491%	17850.5156	17852	0100010101111100	3.34714785	60.1657742	2%	1.265774
	59	3.319930537	-0.000001	0.019500%	0.010395	3.354862732	1.0522%	17898.5152	17902	0100010111011010	3.356522564	60.27827077	2%	1.278271
	59.1	3.329323263	-0.000000	0.019600%	0.010426	3.364458503	1.0553%	17949.7096	17954	0100011000100010	3.366272266	60.39526722	2%	1.295267
	59.2	3.338155747	-0.000000	0.019600%	0.010456	3.373486313	1.0584%	17997.8738	18001	0100011001010001	3.375084497	60.50101397	2%	1.301014
	59.3	3.347003304	-0.000001	0.019700%	0.010487	3.382532312	1.0615%	18046.1351	18049	0100011010000001	3.384084223	60.60901067	2%	1.309011
	59.4	3.355858516	-0.000002	0.019700%	0.010517	3.391590564	1.0645%	18094.4618	18093	0100011010101011	3.392333971	60.70800765	2%	1.308008
	59.5	3.364743165	-0.000002	0.019800%	0.010548	3.400688808	1.0677%	18142.8951	18144	0100011011100000	3.401896179	60.82275415	2%	1.322754
	59.6	3.373635231	-0.000003	0.019800%	0.010578	3.409758038	1.0707%	18191.387	18190	0100011100001110	3.410520916	60.92625099	2%	1.326251
	59.7	3.382541886	-0.000003	0.019900%	0.010609	3.418867041	1.0739%	18239.9844	18242	0100011010000010	3.420270618	61.04324742	2%	1.343247
	59.8	3.392029917	-0.000004	0.019900%	0.010639	3.428359747	1.0769%	18291.6859	18294	0100011011011010	3.430020321	61.16024385	2%	1.360244
	59.9	3.400964358	-0.000004	0.020000%	0.010670	3.437897797	1.0801%	18340.4492	18339	0100011011010011	3.438457963	61.26149076	2%	1.361491
	60	3.4093483	-0.000005	0.020000%	0.010700	3.446275831	1.0831%	18386.2129	18389	0100011110101011	3.447832277	61.3739873	2%	1.373987
	60.1	3.418312177	-0.000006	0.020100%	0.010731	3.455443923	1.0863%	18435.1256	18436	0100100000000100	3.456644508	61.4797341	2%	1.379734
	60.2	3.427290061	-0.000006	0.020100%										

	62.4	3.620018559	0.000019	0.021200%	0.011432	3.661919965	1.1575%	19536.6951	19537	01001100001010001	3.663075708	63.9569085	2%	1.556909	
	62.5	3.628763583	0.000019	0.021300%	0.011463	3.67098123	1.1607%	19584.5043	19585	0100110010000001	3.672075434	64.0649052	3%	1.564905	
	62.6	3.638059215	0.000020	0.021300%	0.011493	3.680395332	1.1637%	19635.263	19636	0100110001010100	3.681637642	64.1796517	3%	1.579652	
	62.7	3.646829647	0.000021	0.021400%	0.011524	3.689382422	1.1668%	19683.2099	19683	01001100011100011	3.690449873	64.28539848	3%	1.585398	
	62.8	3.655613128	0.000021	0.021400%	0.011554	3.698380547	1.1699%	19731.2158	19734	0100110100001010	3.700012081	64.40014497	3%	1.600145	
	62.9	3.663875341	0.000022	0.021500%	0.011585	3.708954578	1.1731%	19776.4256	19778	01001101010000010	3.708261829	64.49914195	3%	1.599142	
	63	3.672685698	0.000022	0.021500%	0.011615	3.719880892	1.1761%	19824.5818	19825	0101010101101100	3.717636543	64.61163852	3%	1.611639	
	63.1	3.681508574	0.000023	0.021600%	0.011646	3.729232366	1.1793%	19872.8242	19875	0100110101000011	3.726448774	64.71738529	3%	1.617386	
	63.2	3.689281952	0.000023	0.021600%	0.011676	3.732901634	1.1823%	19915.3891	19919	01001101011000111	3.734698523	64.81638227	3%	1.616382	
	63.3	3.698132034	0.000024	0.021700%	0.011707	3.741972629	1.1855%	19963.7838	19963	01001101111110111	3.742948271	64.91537925	3%	1.615379	
	63.4	3.707523476	0.000025	0.021700%	0.011737	3.751588186	1.1885%	20015.0837	20016	01001110000110000	3.752885468	65.03462561	3%	1.634626	
	63.5	3.715869121	0.000025	0.021800%	0.011768	3.760150895	1.1917%	20060.7666	20059	0100111000101011	3.760947721	65.13137266	3%	1.631373	
	63.6	3.724228948	0.000026	0.021800%	0.011798	3.768723667	1.1947%	20106.5031	20108	0100111010000100	3.770134941	65.24161929	3%	1.641619	
	63.7	3.733128752	0.000026	0.021900%	0.011829	3.777848232	1.1979%	20155.1836	20160	0100111011000000	3.779884644	65.35861572	3%	1.658616	
	63.8	3.741515555	0.000027	0.021900%	0.011859	3.78644937	1.2010%	20201.0714	20206	01001110110110110	3.78850938	65.46211256	3%	1.662113	
	63.9	3.750440035	0.000027	0.022000%	0.011890	3.795600029	1.2041%	20249.8911	20251	01001111000010111	3.796946623	65.56335947	3%	1.663359	
	64	3.758853343	0.000028	0.022000%	0.011920	3.804229063	1.2072%	20295.9278	20300	01001111010010100	3.806133842	65.67360611	3%	1.673606	
	64.1	3.767280243	0.000029	0.022100%	0.011951	3.812876234	1.2103%	20342.0613	20343	01001111011011011	3.814196096	65.77035316	3%	1.670353	
	64.2	3.776241381	0.000029	0.022100%	0.011981	3.822061808	1.2134%	20391.0672	20391	01001111011001011	3.823195822	65.87834986	3%	1.678350	
	64.3	3.784694082	0.000030	0.022200%	0.012012	3.830736189	1.2165%	20437.3459	20436	01001111101010100	3.831633064	65.97959677	3%	1.679597	
	64.4	3.793678736	0.000030	0.022200%	0.012042	3.839946675	1.2196%	20486.4847	20485	010100000000000101	3.840820284	66.08984341	3%	1.689843	
	64.5	3.802156773	0.000031	0.022300%	0.012073	3.848647798	1.2228%	20532.906	20536	01010000001110000	3.850382492	66.2045899	3%	1.704590	
	64.6	3.810647816	0.000031	0.022300%	0.012103	3.857359739	1.2258%	20579.3851	20584	01010000010100000	3.859382217	66.31258661	3%	1.712587	
	64.7	3.818636104	0.000032	0.022400%	0.012134	3.86556618	1.2290%	20623.1672	20627	01010000010001011	3.867444471	66.40933366	3%	1.709334	
	64.8	3.827153826	0.000033	0.022400%	0.012164	3.874305169	1.2320%	20669.7906	20669	01010000010111010	3.875319231	66.50383077	3%	1.703831	
	64.9	3.835684202	0.000033	0.022500%	0.012195	3.883062641	1.2352%	20716.5115	20715	01010000010101011	3.883943968	66.60732761	3%	1.707328	
	65	3.844227115	0.000034	0.022500%	0.012225	3.891828003	1.2382%	20763.2766	20766	01010000100011100	3.893506176	66.72207411	3%	1.722074	
	65.1	3.852782448	0.000034	0.022600%	0.012256	3.900611633	1.2414%	20810.1381	20812	0101000010001000	3.902130913	66.82557095	3%	1.725571	
	65.2	3.861350085	0.000035	0.022600%	0.012286	3.909403301	1.2445%	20857.0425	20860	01010000101111000	3.911130638	66.93356766	3%	1.733568	
	65.3	3.869929991	0.000035	0.022700%	0.012317	3.918212823	1.2476%	20904.0421	20903	0101000010010011	3.919192892	67.0303147	3%	1.730315	
	65.4	3.878013293	0.000036	0.022700%	0.012347	3.92651526	1.2507%	20948.3364	20948	01010000110010100	3.927630135	67.13156161	3%	1.731562	
	65.5	3.886618181	0.000037	0.022800%	0.012378	3.935350208	1.2538%	20995.4717	20999	01010010000000011	3.937192343	67.24630811	3%	1.746308	
	65.6	3.895234468	0.000037	0.022800%	0.012408	3.944194717	1.2569%	21042.6568	21041	01010010001000000	3.945067102	67.34806523	3%	1.748065	
	65.7	3.903863303	0.000038	0.022900%	0.012438	3.953054629	1.2601%	21089.9285	21093	01010010010010010	3.954818905	67.45780166	3%	1.757802	
	65.8	3.912503348	0.000038	0.022900%	0.012469	3.961923853	1.2631%	21137.2447	21137	01010010100100001	3.963066553	67.55679864	3%	1.756799	
	65.9	3.920651431	0.000039	0.023000%	0.012500	3.970288436	1.2663%	21181.9239	21185	01010010110000001	3.972066278	67.66479534	3%	1.764795	
	66	3.92931536	0.000040	0.023000%	0.012530	3.979191908	1.2693%	21229.3714	21230	0101001011010110	3.980503521	67.76604225	3%	1.766042	
	66.1	3.937489113	0.000040	0.023100%	0.012561	3.987594456	1.2725%	21274.2004	21273	01010010100010001	3.988565775	67.8627893	3%	1.762789	
	66.2	3.945175597	0.000041	0.023100%	0.012591	3.995499197	1.2756%	21316.3724	21320	01010010100010000	3.997378006	67.96836807	3%	1.768536	
	66.3	3.953875995	0.000041	0.023200%	0.012622	4.004436257	1.2788%	21364.0524	21365	01010010101100101	4.005815248	68.06978298	3%	1.769783	
	66.4	3.962587315	0.000042	0.023200%	0.012652	4.013379863	1.2818%	21414.7674	21414	01010010110010100	4.015002468	68.18002962	3%	1.780030	
	66.5	3.970812019	0.000042	0.023300%	0.012683	4.021836232	1.2850%	21456.883	21459	01010010111001010	4.023439711	68.28127653	3%	1.781277	
	66.6	3.979049419	0.000043	0.023300%	0.012713	4.030300897	1.2880%	21502.0428	21506	01010100000000010	4.032251942	68.3870233	3%	1.787023	
	66.7	3.987794823	0.000044	0.023400%	0.012744	4.039284728	1.2912%	21549.9724	21553	0101010000100001	4.041064173	68.49277007	3%	1.792770	
	66.8	3.99605627	0.000044	0.023400%	0.012774	4.047775817	1.2943%	21595.2732	21597	01010100001011011	4.049313921	68.59176705	3%	1.791767	
	66.9	4.004330067	0.000045	0.023500%	0.012805	4.056283028	1.2974%	21640.66	21640	01010100100010000	4.057376175	68.6885141	3%	1.788514	
	67	4.013108526	0.000045	0.023500%	0.012835	4.065298893	1.3005%	21688.7605	21688	01010100101110000	4.0663759	68.7965108	3%	1.796511	
	67.1	4.02140568	0.000046	0.023600%	0.012866	4.073830841	1.3037%	21734.2792	21733	01010100101100101	4.074813143	68.89775719	3%	1.797758	
	67.2	4.029714839	0.000046	0.023600%	0.012896	4.082372365	1.3067%	21779.8491	21780	01010101000010100	4.083625374	69.00350449	3%	1.803504	
	67.3	4.038035887	0.000047	0.023700%	0.012927	4.090929584	1.3099%	21825.5027	21825	01010101010000001	4.092062616	69.1047514	3%	1.804751	
	67.4	4.045880258	0.000048	0.023700%	0.012957	4.099000253	1.3129%	21868.5605	21871	01010101010101111	4.100687353	69.20824824	3%	1.808248	
	67.5	4.054225732	0.000048	0.023800%	0.012988	4.107584283	1.3161%	21914.3571	21919	01010101000110111	4.109687078	69.31624494	3%	1.816245	
	67.6	4.062582751	0.000049	0.023800%	0.013018	4.116175366	1.3192%	21960.1913	21960	01010101010010000	4.117374344	69.40849213	3%	1.808492	
	67.7	4.071436682	0.000049	0.023900%	0.013049	4.125275647	1.3224%	22008.7422	22010	01010101011110100	4.126749058	69.52098869	3%	1.820989	
	67.8	4.078464672	0.000050	0.023900%	0.013079	4.132908017	1.3254%	22049.4616	22054	01010100001001010	4.134998806	69.61998567	3%	1.819986	
	67.9	4.087238427	0.000050	0.024000%	0.013110	4.141541272	1.3286%	22095.5209	22094	01010100100010110	4.142498577	69.70998293	3%	1.809983	
	68	4.095158949	0.000051	0.024000%	0.013140	4.149692142	1.3317%	22139.0066	22143	01010100101111111	4.151685797	69.82022956	3%	1.820230	
	68.1	4.103573951	0.000052	0.024100%	0.013171	4.158348799	1.3348%	22185.1907	22186	01010101010101010	4.159748051	69.91697661	3%	1.816977	
	68.2	4.11151926	0.000052	0.024100%	0.013201	4.166526815	1.3379%	22228.8212	22230	01010101010101010	4.167987799	70.01597359	3%	1.815974	
	68.3	4.120436425	0.000053	0.024200%	0.013232	4.175099343	1.3410%	22272.7259	22278	01010110000000100	4.176997524	70.12387029	3%	1.823970	
	68.4	4.12840508	0.000053	0.024200%	0.013262	4.183999141	1.3441%	22321.4882	22322	01010110010010010	4.185247272	70.22296727	3%	1.822967	
	68.5	4.13686383	0.000054	0.024300%	0.013293	4.192999281	1.3473%	22367.9203	22371	01010110101000111	4				

	70.9	4.330017884	0.000068	0.025500%	0.014025	4.39159925	1.4222%	23429.6042	23432	0101101110001000	4.393365921	72.72039105	3%	1.820391
	71	4.338277002	0.000068	0.025500%	0.014055	4.400109729	1.4253%	23475.0085	23476	0101101110110100	4.401615669	72.81938803	3%	1.819388
	71.1	4.346093091	0.000069	0.025600%	0.014086	4.408174824	1.4284%	23518.0365	23519	0101101110111111	4.409677923	72.91613508	3%	1.816135
	71.2	4.353920518	0.000069	0.025600%	0.014118	4.416248462	1.4315%	23561.1102	23564	0101101000000100	4.418151566	73.01738199	3%	1.817382
	71.3	4.361759177	0.000070	0.025700%	0.014147	4.424337451	1.4347%	23604.2657	23605	0101110000011001	4.425802431	73.10962917	3%	1.809629
	71.4	4.369608939	0.000071	0.025700%	0.014177	4.432433726	1.4378%	23647.4601	23649	0101100001100001	4.434045219	73.20862615	3%	1.808626
	71.5	4.3779178511	0.000071	0.025800%	0.014238	4.441003045	1.4410%	23693.3751	23695	0101100100001111	4.442676916	73.31212298	3%	1.812123
	71.6	4.385789227	0.000072	0.025800%	0.014238	4.449120986	1.4440%	23736.4882	23741	0101100101111101	4.451301653	73.41561983	3%	1.815620
	71.7	4.393223792	0.000072	0.025900%	0.014269	4.456803087	1.4472%	23777.4773	23779	0101100011100011	4.458426435	73.50111723	3%	1.801117
	71.8	4.401116893	0.000073	0.025900%	0.014299	4.464945317	1.4503%	23820.9126	23821	010110100000101	4.466301195	73.59561434	3%	1.795614
	71.9	4.409020552	0.000073	0.026000%	0.014330	4.473104305	1.4535%	23864.4415	23866	0101101001110110	4.474728438	73.69686125	2%	1.796861
	72	4.417378778	0.000074	0.026000%	0.014360	4.481719436	1.4565%	23910.4041	23912	010110101010000	4.483363174	73.80035809	3%	1.800358
	72.1	4.424859096	0.000075	0.026100%	0.014391	4.489448921	1.4597%	23951.6416	23954	010110100100010	4.491237934	73.89485521	2%	1.794855
	72.2	4.432793763	0.000075	0.026100%	0.014421	4.497363634	1.4628%	23995.3223	23997	0101101010111101	4.499300188	73.99160228	2%	1.791602
	72.3	4.440297222	0.000076	0.026200%	0.014452	4.505390284	1.4660%	24036.6903	24041	0101101111101001	4.507549936	74.09059924	2%	1.790599
	72.4	4.448252942	0.000076	0.026200%	0.014482	4.513600058	1.4691%	24080.4903	24082	0101110000010010	4.515237202	74.18284642	2%	1.782846
	72.5	4.455779093	0.000077	0.026300%	0.014513	4.521377801	1.4722%	24121.9864	24123	0101110000110001	4.522549479	74.27059374	2%	1.770594
	72.6	4.463755425	0.000077	0.026300%	0.014543	4.529609696	1.4753%	24165.9032	24170	010111000101010	4.531736698	74.38084038	2%	1.780840
	72.7	4.471303823	0.000078	0.026400%	0.014574	4.537411209	1.4785%	24207.5251	24212	0101110100010100	4.539611458	74.47533749	2%	1.775337
	72.8	4.478863641	0.000079	0.026400%	0.014604	4.545220189	1.4815%	24249.1867	24253	0101110101111011	4.547298723	74.56758468	2%	1.767585
	72.9	4.48687053	0.000079	0.026500%	0.014635	4.553488992	1.4847%	24293.3016	24295	0101110110110011	4.555173483	74.66208179	2%	1.762082
	73	4.494886771	0.000080	0.026500%	0.014665	4.561762162	1.4878%	24337.4397	24342	0101110001010110	4.563985714	74.76782857	2%	1.767829
	73.1	4.502478335	0.000080	0.026600%	0.014696	4.569610478	1.4910%	24379.3113	24379	0101110011011011	4.570923002	74.85107603	2%	1.751076
	73.2	4.510080875	0.000081	0.026600%	0.014726	4.577464767	1.4941%	24421.2146	24422	0101110110100110	4.578985256	74.94782307	2%	1.747823
	73.3	4.518126366	0.000081	0.026700%	0.014757	4.585774789	1.4973%	24465.5494	24467	010111010010011	4.587422499	75.04060998	2%	1.749070
	73.4	4.52574961	0.000082	0.026700%	0.014787	4.593651071	1.5003%	24507.5701	24506	0101110110110110	4.594734776	75.13681731	2%	1.736817
	73.5	4.533383501	0.000083	0.026800%	0.014818	4.60154331	1.5035%	24549.676	24550	0101110111100110	4.602984524	75.23581429	2%	1.735814
	73.6	4.541027928	0.000083	0.026800%	0.014848	4.609443077	1.5066%	24591.822	24596	0100000000010100	4.611960261	75.33931113	2%	1.739311
	73.7	4.548254354	0.000084	0.026900%	0.014879	4.616922684	1.5098%	24631.7264	24630	0100000001011010	4.617984066	75.41580879	2%	1.715809
	73.8	4.556347954	0.000084	0.026900%	0.014909	4.62527938	1.5129%	24676.3102	24679	010000000100111	4.627171286	75.52605543	2%	1.726055
	73.9	4.564023335	0.000085	0.027000%	0.014940	4.633215711	1.5160%	24718.6513	24717	0100000010000101	4.634299068	75.61155282	2%	1.711553
	74	4.570857421	0.000086	0.027000%	0.014970	4.640293975	1.5191%	24756.4133	24759	0100000010101111	4.642170828	75.70604993	2%	1.706050
	74.1	4.578554707	0.000086	0.027100%	0.015001	4.648462343	1.5223%	24798.8838	24803	0100000110000111	4.650504691	75.80594691	2%	1.705047
	74.2	4.586685761	0.000087	0.027100%	0.015031	4.656849998	1.5254%	24843.6755	24848	0100001000100000	4.658857919	75.90629382	2%	1.706294
	74.3	4.593978818	0.000087	0.027200%	0.015062	4.664201142	1.5286%	24883.9615	24885	0100000101001011	4.665795107	75.98954128	2%	1.699541
	74.4	4.601705426	0.000088	0.027200%	0.015092	4.67218172	1.5316%	24926.5679	24929	010000010100001	4.674044855	76.08853826	2%	1.688538
	74.5	4.609020417	0.000088	0.027300%	0.015123	4.679761574	1.5348%	24966.9779	24966	010000100000110	4.680962143	76.17178572	2%	1.671786
	74.6	4.616769637	0.000089	0.027300%	0.015153	4.687768819	1.5379%	25009.6974	25011	0100000101100011	4.689419396	76.27303263	2%	1.673033
	74.7	4.624522802	0.000090	0.027400%	0.015184	4.695790809	1.5411%	25052.4955	25057	010000011000001	4.698044123	76.37652947	2%	1.676529
	74.8	4.631869424	0.000090	0.027400%	0.015214	4.703393978	1.5442%	25093.0591	25098	01000010000001010	4.705731388	76.46877668	2%	1.668777
	74.9	4.639222671	0.000091	0.027500%	0.015245	4.7101012472	1.5474%	25133.7045	25137	01000010000100001	4.713043665	76.55652398	2%	1.656524
	75	4.647012023	0.000091	0.027500%	0.015275	4.719061806	1.5505%	25176.6485	25181	01000010010110101	4.721293413	76.65552096	2%	1.655521
	75.1	4.654390197	0.000092	0.027600%	0.015306	4.726702214	1.5536%	25217.4108	25217	0100001000000001	4.728043207	76.73651849	2%	1.636518
	75.2	4.66177897	0.000092	0.027600%	0.015336	4.734350073	1.5567%	25258.2128	25258	0100001010101010	4.735730473	76.82876567	2%	1.628766
	75.3	4.669178231	0.000093	0.027700%	0.015367	4.742012851	1.5599%	25299.0945	25301	0100001010101011	4.743792727	76.92551272	2%	1.625513
	75.4	4.677000998	0.000094	0.027700%	0.015397	4.750101417	1.5630%	25342.2478	25342	0100001011111110	4.751479992	77.0177599	2%	1.617760
	75.5	4.684420025	0.000094	0.027800%	0.015428	4.757786256	1.5662%	25383.2471	25383	0100001001000111	4.759167257	77.11000799	2%	1.610007
	75.6	4.691849218	0.000095	0.027800%	0.015458	4.76547605	1.5692%	25424.2729	25428	0100001010101000	4.7676045	77.211254	2%	1.611254
	75.7	4.698878004	0.000095	0.027900%	0.015489	4.77276547	1.5724%	25463.1627	25465	0100001010111001	4.774541788	77.29450146	2%	1.594501
	75.8	4.706328091	0.000096	0.027900%	0.015519	4.78047742	1.5755%	25504.3067	25506	0100001010000010	4.782229053	77.38674864	2%	1.586749
	75.9	4.713788021	0.000096	0.028000%	0.015550	4.788205718	1.5787%	25545.5379	25546	01000011000101010	4.789728825	77.4767459	2%	1.576746
	76	4.720849869	0.000097	0.028000%	0.015580	4.795524233	1.5818%	25584.5829	25589	01000011101110011	4.797791079	77.57349294	2%	1.573493
	76.1	4.728736985	0.000098	0.028100%	0.015611	4.803686428	1.5850%	25628.129	25631	0100100000011111	4.805665838	77.66799006	2%	1.567990
	76.2	4.736225807	0.000098	0.028100%	0.015641	4.811440643	1.5881%	25669.4984	25669	01001000010000101	4.812790621	77.75348745	2%	1.553487
	76.3	4.743318862	0.000099	0.028200%	0.015672	4.818797142	1.5913%	25708.7461	25707	01000010000101011	4.819915403	77.83898484	2%	1.538985
	76.4	4.750422983	0.000099	0.028200%	0.015702	4.826161459	1.5944%	25748.0354	25753	0100001000010001	4.82854014	77.94248168	2%	1.542482
	76.5	4.75834493	0.000100	0.028300%	0.015733	4.834361019	1.5975%	25791.7808	25794	0100000100000010	4.836227406	78.03472887	2%	1.534729
	76.6	4.765469106	0.000100	0.028300%	0.015763	4.841746544	1.6006%	25831.1839	25834	01000001010101010	4.843727177	78.12472612	2%	1.524726
	76.7	4.772604022	0.000101	0.028400%	0.015794	4.849147542	1.6038%	25870.6684	25870	010001000001110	4.850476971	78.20572365	2%	1.505724
	76.8	4.780150394	0.000102	0.028400%	0.015824	4.856906204	1.6069%	25912.3595	25911	0100010010010111	4.858164236	78.29797083	2%	1.497971
	76.9	4.787305599	0.000102	0.028500%	0.015855	4.864385512	1.6101%	25951.9644	25954	0100010101000010	4.866272649	78.39471788	2%	1.494718
	77	4.794471223	0.000103	0.028500%	0.015885	4.871814968	1.6132%	25991.5965	25991	0100010100000111	4.873163778	78.47796534	2%	1.477965
	77.1	4.801647157	0.000103	0.028600%	0.015916	4.879299529	1.6164%	26031.3187	26031	0100010101011111	4.880663549	78.56796259	2%	1.467963
	77.2	4.809230655	0.000104											

79.4	4.956889368	0.000117	0.029700	0.016617	<u>5.049740949</u>	1.6886%	26940.8535	26940	01101001001111100	5.051095849	80.61315018	2%	1.213150
79.5	4.972331075	0.000117	0.029800	0.016648	<u>5.058654236</u>	1.6918%	26977.7482	26977	011010011000011	5.058603136	80.69639764	2%	1.196398
79.6	4.979565176	0.000118	0.029800	0.016678	<u>5.063908047</u>	1.6948%	27016.718	27018	0110100110001010	5.065720402	80.78864482	1%	1.188645
79.7	4.986606256	0.000118	0.029900	0.016709	<u>5.071281327</u>	1.6981%	27055.735	27056	01101001101110000	5.072845184	80.87414221	1%	1.174242
79.8	4.993656614	0.000119	0.029900	0.016739	<u>5.078605414</u>	1.7011%	27094.8482	27096	01101001101110100	5.080344965	80.96413495	1%	1.164139
79.9	5.000341479	0.000119	0.030000	0.016770	<u>5.085644422</u>	1.7043%	27131.9752	27137	01101010000000001	5.088032221	81.05638665	1%	1.156387
80	5.002267689	0.000120	0.030000	0.016800	<u>5.08767772</u>	1.7074%	27143.2497	27143	01101010000000111	5.089157187	81.06986244	1%	1.069886